

Name: Archana A D

## Title: Data Analysis with Pandas

### Aim

To perform data analysis on a CSV dataset using the Pandas library.

### Software Requirements

- Python 3.12.5
- Windows OS
- VS Code / Notepad
- Pandas Library

### Program

```
import pandas as pd # Import pandas library

try:
    # Read CSV file
    df = pd.read_csv("students.csv")

    # Display complete dataset
    print("Student Dataset")
    print(df)

    # Calculate average marks
    print("\nAverage Marks:")
    print(df["Marks"].mean())

    # Filter CSE students
    print("\nCSE Students:")
    print(df[df["Department"] == "CSE"])

    # Department-wise average marks
    print("\nDepartment Wise Average Marks:")
    print(df.groupby("Department")["Marks"].mean())

except Exception as e:
    print("Error:", e)
```

### Explanation

- Imports the Pandas library for data analysis.
- Reads data from a CSV file using read\_csv().
- Displays the complete dataset.

- Calculates the average marks using mean().
- Filters students belonging to the CSE department.
- Groups data department-wise and calculates average marks.
- Uses exception handling to manage errors.

## Output

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VICTUS\OneDrive\Desktop\Task3_Data_Analysis>python analysis.py
Student Dataset
   Name Department  Marks
0  Archana         CSE    90
1   Rahul         ECE    85
2    Anu          CSE    88
3  Akhil          ME    75
4   Meera         CSE    95

Average Marks:
86.6

CSE Students:
   Name Department  Marks
0  Archana         CSE    90
2    Anu          CSE    88
4   Meera         CSE    95

Department Wise Average Marks:
Department
CSE    91.0
ECE    85.0
ME     75.0
Name: Marks, dtype: float64
```

## Conclusion

The program successfully analyzed student data using Pandas, calculated averages, filtered records, and generated department-wise insights.